# D2 sleep-stage criteria
# An empirical reconstruction

10-page technical report | D2 data only
Supplied backup and D2 importer source | 6 November 2026

**Result:** the export supports decoding D2 labels, measuring their associations and testing candidate rules. It does **not** identify the exact proprietary classifier. A simple awake/HR association survives testing; light, deep and REM are poorly separated by the exported parameters.

## What was actually analyzed

Only the d2 branch of the backup was used: imported watch sleep-stage sequences, activity minutes and watch HRV records. No chest-strap data, synchronized comparison results or another device's classifier contributed to any model or inference. Demo data were excluded; no synthetic records were generated.

| Evidence | Amount |
|---|---|
| D2 sleep records | 94 records from one LCA-B10 watch / one user |
| Stage-labelled minutes | 23,221 specific-stage minutes; 2,056 generic nap minutes |
| Physiology modelling cohort | 3,791 minutes with HR or recent D2 HRV |
| Validation | 19 held-out recording-end-date groups |
| Strongest simple association | HR >=55 bpm versus D2 awake: binary balanced accuracy 73.6%, but positive predictive value only 21.8% |

## How to read the conclusions

**Verified facts** are code meanings and exported sequence properties. **Empirical rules** are fitted associations, checked against unseen recording days. **Unresolved criteria** include internal motion thresholds, pulse-wave features, breathing coherence, normalization, temporal smoothing and model weights. None is silently replaced by a guessed value.

The target is fidelity to D2 labels, not clinical accuracy. There is no reference EEG sleep scoring in this analysis. Neither a good nor a poor match to D2 proves the biological stage.

# 01 Available D2 evidence

The labels are dense; the corresponding exported parameters are sparse.

| D2 label | Code | Minutes across all records |
|---|---|---|
| Light sleep | 1 | 11,830 |
| REM | 2 | 4,530 |
| Deep sleep | 3 | 5,214 |
| Awake | 4 | 1,647 |
| Generic nap sleep | 5 | 2,056 |

## Scope and provenance

The 94 stored D2 records span September 2025 through November 2026, with a long gap between periods. There are 57 records containing specific sleep stages and 37 generic nap records. Their 25,277 stage minutes have no overlapping timestamps or conflicting labels in this export. Physiological models use a more recent subset, not every labelled minute.

| Exported feature aligned to specific stages | Available minutes | % of 23,221 |
|---|---|---|
| Heart rate, exact clock minute | 658 | 2.8% |
| D2 HRV, latest within 5 minutes | 3,299 | 14.2% |
| SpO2, exact minute | 298 | 1.3% |
| Step count, exact minute | 0 | 0.0% |
| Energy field, exact minute | 47 | 0.2% |

Activity records cover 4,210 of the specific-stage minutes across 11 sleep records. Most fields within those rows are still null. A stored minute row is therefore not evidence of a valid HR, motion or oxygen measurement. In particular, absent steps cannot be replaced with zero or interpreted as stillness.

There are 662 HRV reading timestamps falling on specific-stage minutes; the five-minute association expands their coverage to 3,299 minutes. Those repeated associations are not 3,299 independent HRV measurements. Raw beat intervals, PPG waveforms, accelerometer vectors and a continuous breathing series are not available in the analyzed D2 export.

Sources: backup d2.nights and d2.records; D2 source src/d2/extra.ts, records.ts and sleepfile.ts. Field 500044831 is labelled RMSSD (ms) by the supplied importer; its actual firmware calculation window is not established.

# 02 Reconstruction method

A reproducible label-to-parameter join, followed by tests on held-out days.

| Step | Method |
|---|---|
| Stage timing | One exported code per minute from each D2 record's start. Full stored sequences are retained, including awake boundaries. |
| Activity join | Match Unix clock-minute slots. HR, oxygen and energy remain missing when absent; no HR interpolation. |
| HRV association | Latest D2 HRV timestamp not later than the end of the stage minute, and no older than 300 seconds relative to that end. |
| HR features | Exact-minute HR; trailing five-minute HR mean and sample SD if >=3 observations; change from the immediately preceding minute when both exist. |
| Training rows | Codes 1-4, with at least observed HR or associated HRV. Code 5 is never relabelled as light. |
| Imputation | Training-fold medians only, plus a separate missingness indicator for every feature. All-missing fields contribute no real information. |
| Validation unit | Leave one recording-end-date out, refit on the other 18 groups, and pool held-out predictions. No random minute split. |

## Model families tested

A majority-light baseline is compared with depth-3 classification trees using a minimum of 60 training rows per leaf. The physiological features are HR, HRV, SpO2, steps, energy, trailing HR mean/SD and HR change. Class-balanced trees additionally give each class equal total training weight. Context variants add clock time and elapsed minutes from the exported record start.

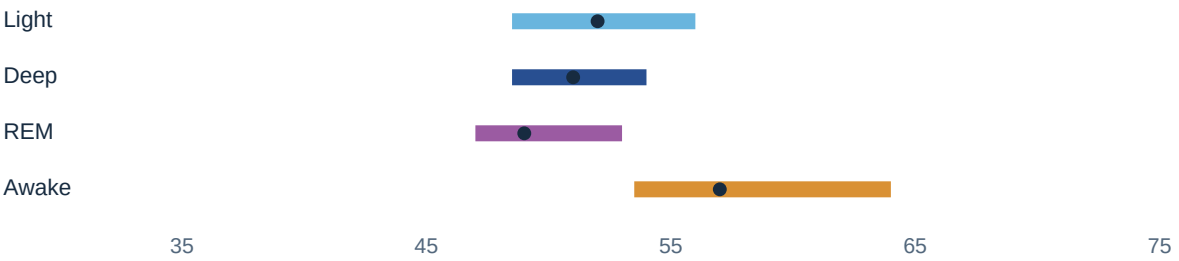
Elapsed time from an exported sleep record is an **output-dependent retrospective context feature**, not an independently observed firmware input. It can exploit how records were segmented. Context results are therefore reported separately and do not establish a real-time decision rule.

No whole-night averages, sleep scores, stage percentages, true previous labels or final-night duration were copied onto minute rows as predictors. This avoids obvious target leakage. Sparse physiological sampling and repeated HRV snapshots still limit identifiability.

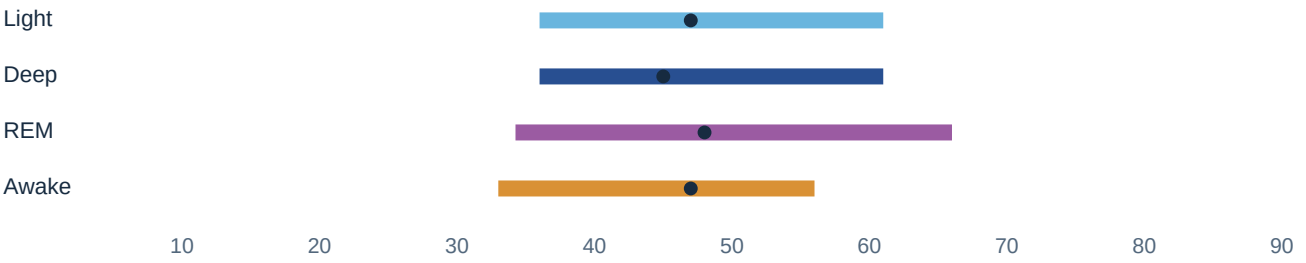
# 03 Do the parameters separate stages?

Substantial overlap contradicts a simple universal HR or HRV lookup table.

**Heart rate: median and middle 50% (bpm)**




**Associated HRV: median and middle 50% (ms)**




| Stage | HR n; median [Q1-Q3] | HRV-associated n; median [Q1-Q3] |
|---|---|---|
| Light | 363; 52 [48.5-56] | 1,810; 47 [36-61] |
| Deep | 111; 51 [48.5-54] | 666; 45 [36-61] |
| REM | 125; 49 [47-53] | 670; 48 [34.25-66] |
| Awake | 59; 57 [53.5-64] | 153; 47 [33-56] |

## Direct counterexamples

At exactly **49 bpm**, D2 assigns 35 light minutes, 19 REM minutes, 10 deep minutes and 3 awake minutes. At an associated HRV of **60 ms**, there are 25 light, 5 REM and 10 deep minutes. The same exported value therefore does not uniquely determine a label.

Median REM HR is lower than median deep HR in the observed samples. This does not invert sleep physiology; it shows that sparse downstream HR readings cannot be treated as the classifier's complete input. The distributions also pool nights with different baselines. A deep-labelled HR outlier of 162 bpm is retained under the importer's validity rule, not accepted as proof that the firmware used that value.

Intervals show the interquartile range, not a confidence interval or manufacturer threshold. HR and HRV columns have different sample counts and availability patterns; they are not paired for every row.

# 04 Candidate numerical thresholds

These are fitted one-stage-versus-rest rules, not four mutually exclusive firmware criteria.

| Target / input | Full-data candidate rule | Held-out binary BA |
|---|---|---|
| Awake / HR | HR > 54.5 | 73.6% |
| Awake / HRV | HRV <= 56.5 | 41.7% |
| Light / HR | HR > 53.5 | 46.0% |
| Light / HRV | HRV <= 67.5 | 53.3% |
| Deep / HR | HR <= 54.5 | 54.2% |
| Deep / HRV | HRV <= 39.5 | 50.4% |
| REM / HR | HR <= 53.5 | 58.0% |
| REM / HRV | HRV > 67.5 | 51.7% |

## How the thresholds were selected

For each target and feature, every distinct observed midpoint and both inequality directions were tested. The chosen rule maximizes binary balanced accuracy on the training data. The held-out column repeats that selection within each recording-day fold, then pools predictions. HR tests use 658 observed-HR minutes; HRV tests use 3,299 associated-HRV minutes. A 50% binary BA is the non-discriminating reference.

## Awake is the strongest, but not a usable stand-alone stage rule

The fitted HR rule is **HR >54.5 bpm**, equivalent to at least 55 bpm for integer HR. Against the D2 awake label, held-out sensitivity is 72.9% and specificity 74.3%. However, only 43 of 197 HR-high minutes are D2-awake: precision is **21.8%**. There are 154 sleep-labelled false alarms.

Removing pre-sleep/post-sleep awake boundary minutes leaves 30 internal awake HR observations. The same threshold identifies 18 (60.0%), while 154 sleep-labelled false alarms remain. Precision falls to 10.5%. Thus much of the apparent awake separation reflects boundary behavior; it does not recover a reliable internal-awakening detector.

The other thresholds are weak, inconsistent or conflicting. HR <=53.5 can favor REM, while HR <=54.5 can favor deep. Those overlapping statements cannot classify four states without additional information and arbitration. The poor HRV results do not support any firm HRV cutoff.

BA = (sensitivity + specificity) / 2 for one target versus all other specific stages. No threshold was selected using a random split of adjacent minutes. Full-data rule values are illustrative refits; validation evaluates the rule-fitting procedure.

# 05 Temporal clues in D2 labels

Duration and transition patterns are more recoverable than hidden sensor thresholds.

| Label | Number of bouts | Shortest | Median | Longest |
|---|---|---|---|---|
| Light | 525 | 3 min | 16 min | 102 min |
| Deep | 301 | 3 min | 13 min | 68 min |
| REM | 300 | 3 min | 12 min | 51 min |
| Awake | 165 | 1 min | 5 min | 204 min |
| Generic nap | 37 | 3 min | 46 min | 129 min |

## What these patterns suggest

All observed light/deep/REM bouts are at least 3 minutes, whereas awake bouts can last 1 minute. This is consistent with temporal smoothing, a minimum-bout policy or compressed output reporting. It does not distinguish among them or establish an exact three-minute firmware constraint. The supplied parser expands exported durations; it does not impose a three-minute minimum.

| Stage change | To light | To REM | To deep | To awake |
|---|---|---|---|---|
| From light | - | 206 | 217 | 57 |
| From REM | 195 | - | 82 | 21 |
| From deep | 179 | 92 | - | 30 |
| From awake | 151 | 2 | 2 | - |

Counts are changes between adjacent runs within each record. Self-transitions are omitted and no transitions are created across record boundaries.

Light commonly mediates entry from awake: 151 of 155 observed awake-to-specific-sleep transitions enter light. Yet exported labels also contain direct deep/REM transitions, so a compulsory light-between-every-stage rule would fail to reproduce the data.

Among 57 specifically staged records, the first REM label appears a median 42 minutes after the first sleep label; the minimum is 7 minutes, and 34 records reach REM within 50 minutes. First deep appears a median 35 minutes after the first sleep label, with a minimum of 7 minutes. These are exported-record timings, not verified biological REM latencies: truncated sessions or prior sleep can change the apparent starting point.

Inference strength: strong for observed sequence properties; tentative for any hidden temporal implementation. Nap code 5 provides no light/deep/REM training target.

# 06 An explicit surrogate rule set

A fitted decision tree shows what can be approximated, and why it is insufficient.

**Illustrative full-cohort refit of the class-balanced physiological tree**

1. If imputed HR >54.5 bpm: predict **awake**.
2. Otherwise, if a trailing five-minute HR mean is available: predict **awake**.
3. Otherwise, if associated/imputed HRV >79.58 ms: predict **REM**.
4. Otherwise: predict **deep**.

**This particular refit has no light-predicting leaf.** It is an unsuccessful surrogate, not a recovered Huawei algorithm.

## Missing values are central to this tree

In the full-cohort refit, missing HR is imputed as 51 bpm and missing HRV as 47 ms. The model also receives missingness indicators. Consequently, an HR branch can route many rows that have no actual HR measurement. Availability-based branches can reflect the export/sampling process rather than physiology.

The rule set above simplifies branches whose final predicted labels are identical. The HRV split is a numerical CART search result; 79.58 ms is not a meaningful physiological precision or a manufacturer setting. With integer HRV values, it separates 79 from 80 ms. Grouped validation refits the tree each time, so individual fold trees need not have these exact branches.

## Class balancing does not make a rule clinically plausible

Class balancing gives rare awake or deep labels more influence. In the large deep-predicting leaf, the original labels are 1,797 light, 661 REM, 687 deep and 138 awake minutes. It predicts deep because of reweighting, despite light being the raw majority. That illustrates why a tree's printed label is not evidence of a true decision criterion.

The unweighted physiological tree mostly defaults to light, while balanced variants sacrifice overall fidelity to identify more minority labels. Neither behavior supports a faithful four-stage reconstruction. The useful output is the tested limitation and the few modest associations, not a production-ready classifier.

Surrogate specification: maximum depth 3, minimum 60 training rows per leaf, deterministic Gini search with at most 80 candidate cuts per feature/node; training-only imputation. This explicit rule set was fitted only from D2-derived data.

# 07 Held-out validation

No tested four-stage surrogate provides adequate fidelity to D2 labels.

| Held-out model | Accuracy | 4-class balanced accuracy | Macro F1 |
|---|---|---|---|
| Always light | 54.8% | 25.0% | 0.177 |
| Physiology tree | 54.3% | 25.2% | 0.187 |
| Physiology + context | 55.7% | 31.3% | 0.280 |
| Balanced physiology | 22.8% | 28.0% | 0.203 |
| Balanced + context | 28.3% | 32.8% | 0.254 |

## Class-balanced physiological model: pooled confusion matrix

| Actual D2 / prediction | Light | REM | Deep | Awake |
|---|---|---|---|---|
| Light | 233 | 314 | 1337 | 192 |
| REM | 91 | 106 | 513 | 58 |
| Deep | 90 | 123 | 482 | 57 |
| Awake | 16 | 11 | 123 | 45 |

3,791 held-out minute predictions; each date was unseen during its model fit. Counts are D2-label fidelity, not clinical accuracy.

The majority-light baseline achieves 54.8% accuracy because light is common. The ordinary physiological tree reaches only 54.3%, with 25.2% four-class balanced accuracy. Adding clock/elapsed context produces 55.7% accuracy and 31.3% balanced accuracy: a small overall gain that can exploit record segmentation rather than sensor criteria.

The balanced physiological model has 22.8% accuracy and 28.0% balanced accuracy. Its class recalls are 11.2% light, 13.8% REM, 64.1% deep and 23.1% awake. The context-balanced model has 32.8% balanced accuracy but only 28.3% overall accuracy. These tradeoffs are not acceptable evidence of a recovered classifier.

## Why high minute counts do not solve the problem

Labels persist in multi-minute bouts and a single HRV reading may be reused for several adjacent rows. There are 19 independent recording-date groups in the modelling cohort, all from one watch/user. Grouped testing reduces adjacent-minute leakage, but it cannot establish performance on new users, firmware versions or truly unseen sensor distributions.

Balanced accuracy is mean per-class recall. A 25% four-class BA is the non-discriminating reference; macro F1 weights classes equally. No superiority confidence interval is claimed.

# 08 Reconstructed criteria: confidence

Separate verified export behavior from hypotheses about internal classification.

| Claim / criterion | Evidence | Assessment |
|---|---|---|
| Codes 1/2/3/4 map to light/REM/deep/awake | D2 stage parser and stage mapping; exported records | Verified decoding |
| Code 5 means unstaged nap sleep | 37 records; no specific stages in those naps | Verified output semantics |
| Higher HR tends to favor awake | Awake median 57 bpm; held-out HR stump BA 73.6% | Moderate association; poor precision |
| Light has a distinct HR/HRV band | Broad overlapping distributions; weak stump results | Not supported |
| Deep is selected by one low-HR or high-HRV cutoff | Deep/REM/light overlap and weak held-out discrimination | Not supported |
| REM is selected by a fixed HRV or HR threshold | Only modest low-HR association; full models fail | Not supported |
| Sleep-stage bouts are smoothed to >=3 minutes | No observed light/deep/REM bout under 3 minutes | Plausible mechanism; not proven |
| Exact motion, breathing or neural-feature thresholds | Required raw inputs / model internals are absent | Not identifiable here |

## What “not identifiable” means

Several different internal models can generate the same exported label sequence and the same sparse summaries. They may use different PPG features, movement thresholds, respiratory coupling, individual baselines or temporal priors. Downsampled HR and periodic HRV do not uniquely determine which model Huawei used.

Huawei's public documentation describes motion and physiological features and, for earlier TruSleep generations, cardiorespiratory coupling. It does not publish the full D2 decision tree or weights. The supplied backup has no reliable active sleep-algorithm version. Family-level descriptions therefore guide hypotheses but cannot confirm this device's exact criteria. [1]

The supplied D2 Link source decodes watch results. It is not Huawei firmware and contains no implementation of the proprietary four-stage classifier. Output-pattern inference should not be mistaken for code recovery.

# 09 Next reverse-engineering steps

A concrete route to stronger criteria, plus the evidence trail.

## What to collect next

Preserve timestamped D2 stage outputs together with the closest available D2-native PPG/beat intervals, accelerometer movement, respiration and signal-quality flags. Keep exact sample timestamps and firmware/TruSleep versions. Record data at their native rate where the device permits; sparse displayed summaries cannot reconstruct features that were never exported.

For the current bridge, first validate the semantics of the 12-element stress feature vector rather than assigning physiological names by guess. There are 151 stress-record start timestamps, 53 emotion timestamps and 44 blood-pressure timestamps on specific-stage minutes. These were inventoried but excluded from the interpretable physiological model; they are separate downstream readouts and their role in sleep classification is unproven.

## How to test the next candidate

Use whole unseen nights and then additional users. Compare native-signal models against time-only and missingness-only baselines, check firmware changes, report per-stage precision/recall and transition errors, and test whether candidate thresholds remain stable across nights. Analyze exact-timestamp and alternative HRV association windows separately. Label any successful model a surrogate until internal implementation evidence is available.

## Reproducibility and limits

Input: /Users/mike/Downloads/openbeat-backup-2026-11-06-10-57.json
SHA-256: bc38d8055ffe5bf324c814479673407f7d1fae3e8789b1d6ed94a47323519b91
Only d2.nights and d2.records enter the analysis. Models use 3,791 eligible rows and 19 date groups, 19 Oct-6 Nov 2026. No whole-night averages or another device's parameters were used.

Local source [2]: src/d2/sleepfile.ts (parseStages / parseSummary); src/d2/extra.ts (activity and HRV fields); src/d2/records.ts (record formats); src/d2/night.ts (stage totals / nap semantics). Source root:
/Users/mike/Desktop/01 Elitebook 13 Apr 2026/Documents/programming/ECG Analysis/GitHub uploads/2026-11-06 0001 publish/

[1] Huawei: Introduction to sleep monitoring principles. Accessed 6 November 2026. Manufacturer-level context only.

**Final finding:** an awake-related HR association and several output-sequence properties can be reconstructed. The export does not support credible numeric criteria that reliably distinguish all four D2 sleep stages. A validated surrogate requires richer D2-native signals; claiming exact firmware thresholds now would exceed the evidence.